

Phase5 Daily Schedule

Phase 5 — Day-by-Day Schedule · Molecular-ML / Drug-Discovery Track (5A-5F)

Weeks 43-57 · Mon Jun 28 → Sun Oct 10, 2027 · ~450 hrs · → Tier 2 (the DESRES / Isomorphic differentiator)

Goal: across public data — property prediction (fingerprint→D-MPNN→3D-equivariant), molecular foundation models, generative chemistry (GFlowNets/diffusion), the physics stack (conformer→docking→FEP→MD), calibrated UQ, and applied cheminformatics. Every week ends in something you build and can defend cold.

*Blocks: **A** 06-08 (theory) · **B** 08:30-10:30 (build + threads) · **Evening** 20:30-22:30 (papers, Anki, R). Weekend = buffer + rest (this phase is dense — use it). Threads: **T** DSA (Block B ~45m) · **M** implement (Block B) · **R** research/apply (Evening).*

5A — Geometric & molecular ML core (Weeks 43-45)

Week 43 · Jun 28-Jul 4 — Graph ML foundations + molecular representations			
Day	Block A	Block B (T/M)	Evening (+ R)
Mon	CS224W: node/graph embeddings	T: graphs → PyG GCN on a toy set	Anki
Tue	Message passing; GNN layers	M: a message-passing layer from scratch	Anki
Wed	GNN expressiveness (Weisfeiler-Lehman)	T: DP → GraphSAGE/GAT	Anki
Thu	Molecular representations: fingerprints/descriptors	M: RDKit Morgan + descriptors featurizer	R: pick MLSB-track paper
Fri	Scaffold splits (the honest split)	T: timed set	Whiteboard-Fri: message-passing update + permutation invariance
Sat-Sun	Buffer + rest · <i>Deliverable:</i> baselines (fingerprint+GBM) on a TDC ADMET task		
Week 44 · Jul 5-11 — Message-passing for molecules (D-MPNN / Chemprop)			
Day	Block A	Block B	Evening
Mon	Chemprop / D-MPNN paper	M: train Chemprop on your TDC task	Anki
Tue	Directed messages; tottering	T: graphs → ablate directed vs undirected	Anki
Wed	Feature concatenation; readouts	M: finish from-scratch MPNN; beat baseline	Anki
Thu	Graph transformers (Graphormer/GPS)	T: DP → run a graph-transformer demo	R: reproduce-paper
Fri	Over-smoothing + fixes	T: timed set	Whiteboard-Fri: derive D-MPNN edge update
Sat-Sun	Buffer + rest		
Week 45 · Jul 12-18 — 3D / equivariant nets + 🏆 5A capstone			
Day	Block A	Block B	Evening
Mon	Geometric DL (Bronstein): equivariance	M: verify equivariance empirically	Anki
Tue	SchNet → DimeNet++	T: graphs → run SchNet on QM9	Anki
Wed	EGNN (E(n)-equivariance)	M: an EGNN layer; QM9 property	Anki
Thu	NequIP / MACE (potentials)	T: DP → data-efficiency ablation	R: write-up
Fri	2D vs 3D — when each wins	Build the representation ladder (fp vs D-MPNN vs 3D)	Whiteboard-Fri: invariance vs equivariance
Sat-Sun	Buffer + rest · <i>Deliverable:</i> 5A capstone (representation ladder, public repo)		

5B — Molecular foundation models & SSL (Weeks 46-47)

Week 46 · Jul 19-25 — SSL objectives + SMILES/graph pretraining			
Day	Block A	Block B	Evening
Mon	ChemBERTa (masked-SMILES)	M: tokenize + mask a SMILES set	Anki
Tue	MolCLR (contrastive)	T: graphs → a contrastive augmentation	Anki
Wed	Grover / Uni-Mol (3D-aware)	M: small masked-SMILES pretrain run	Anki
Thu	Valid molecular augmentations	T: DP → augmentation checks	R: reproduce-paper
Fri	What each SSL signal captures	T: timed set	Whiteboard-Fri: the MLM objective
Sat-Sun	Buffer + rest		

Week 47 · Jul 26-Aug 1 — Finetune + transfer study (🎯 5B capstone)			
Day	Block A	Block B	Evening
Mon	Fine-tuning strategies	M: finetune a pretrained encoder on a TDC task	Anki
Tue	Linear-probe vs full finetune	T: graphs → linear-probe baseline	Anki
Wed	Transfer at 100/1k/10k labels	M: data-ablation study	Anki
Thu	When SSL helps vs hurts	T: DP → compare vs from-scratch (5A)	R: write-up
Fri	Wrap 5B capstone	Polish repo + transfer-gain plot	Whiteboard-Fri: when a fingerprint+GBM still wins
Sat-Sun	Buffer + rest · <i>Deliverable:</i> 5B capstone (pretrain→finetune transfer gain)		

5C — Generative chemistry (Weeks 48-49)

Week 48 · Aug 2-8 — Generative baselines + landscape + metrics			
Day	Block A	Block B	Evening
Mon	VAE/flow baselines (JT-VAE, MoFlow)	M: a small molecular VAE	Anki
Tue	Generative landscape for molecules	T: graphs → validity/novelty metrics	Anki
Wed	Metrics: validity/unique/novel/diverse	M: a metrics + reward module (QED/SA)	Anki
Thu	Reward design (property proxies)	T: DP → reward function	R: reproduce-paper
Fri	MOSES/GuacaMol benchmarks	T: timed set	Whiteboard-Fri: why high-reward alone → mode collapse
Sat-Sun	Buffer + rest		

Week 49 · Aug 9-15 — GFlowNets + equivariant diffusion (🎯 5C capstone)			
Day	Block A	Block B	Evening
Mon	GFlowNet theory (flow-matching / TB)	M: GFlowNet tutorial env	Anki
Tue	Reward- α -diversity guarantee	T: graphs → GFlowNet on a fragment env	Anki
Wed	Equivariant diffusion (EDM/GeoDiff)	M: run a small 3D generation	Anki
Thu	Diffusion vs GFlowNet vs RL	T: DP → evaluate diversity/validity	R: write-up
Fri	Wrap 5C capstone	Polish generator + evaluation	Whiteboard-Fri: the GFlowNet objective (reward-proportional diversity)
Sat-Sun	Buffer + rest · <i>Deliverable:</i> 5C capstone (property-reward generator, public repo)		

5D — Physics-based structure & simulation (Weeks 50-53)

Week 50 · Aug 16-22 — Conformer generation & 3D analysis			
Day	Block A	Block B	Evening

Mon	RDKit ETKDgv3; MMFF94s	M: generate + optimize conformers	Anki
Tue	Butina RMSD clustering	T: DP → cluster ensembles	Anki
Wed	COV/MAT vs GEOM-Drugs	M: COV/MAT/RMSD benchmark	Anki
Thu	DFT re-ranking (Psi4/PySCF) — your strength	T: run DFT single points on top-k	R: reproduce-paper
Fri	MMFF-vs-DFT ranking correlation	T: timed set	Whiteboard-Fri: why MMFF ordering is unreliable; COV vs MAT
Sat-Sun	Buffer + rest · <i>(follows the conformer capstone spec)</i>		

Week 51 · Aug 23-29 — Molecular docking			
Day	Block A	Block B	Evening
Mon	AutoDock Vina: search + scoring	M: dock a public target/ligand set	Anki
Tue	Enrichment (EF/BEDROC), pose RMSD	T: DP → compute enrichment	Anki
Wed	gnina (CNN scoring)	M: re-score with gnina; compare	Anki
Thu	DiffDock (generative docking)	T: graphs → run DiffDock	R: write-up
Fri	What a docking score really approximates	T: timed set	Whiteboard-Fri: pose RMSD vs enrichment
Sat-Sun	Buffer + rest		

Week 52 · Aug 30-Sep 5 — Free energy (FEP / ABFEP)			
Day	Block A	Block B	Evening
Mon	Stat-mech: free energy, ensembles	M: set up a toy alchemical transform	Anki
Tue	Thermodynamic cycles; error cancellation	T: DP → cycle bookkeeping	Anki
Wed	OpenFE: relative FEP setup	M: a small relative-FEP calc	Anki
Thu	Reading FEP results critically	T: graphs → analyze output	R: reproduce-paper
Fri	When FEP is worth the cost	T: timed set	Whiteboard-Fri: the thermodynamic cycle behind relative FEP
Sat-Sun	Buffer + rest		

Week 53 · Sep 6-12 — Molecular dynamics + 🏆 5D capstone			
Day	Block A	Block B	Evening
Mon	OpenMM: force fields, integrators	M: a short protein–ligand MD	Anki
Tue	Thermostats/barostats; PBC	T: DP → set up an NVT/NPT run	Anki
Wed	MDAnalysis (RMSD/RMSF/contacts)	M: trajectory analysis	Anki
Thu	ML potentials (MACE-OFF/ANI) as surrogates	T: re-score with an ML potential	R: write-up
Fri	The full accuracy/cost ladder	Wrap 5D capstone (conformer→dock→MD + FEP)	Whiteboard-Fri: the ladder + where each breaks (ties to your VQE work)
Sat-Sun	Buffer + rest · <i>Deliverable:</i> 5D capstone (public target; pose/enrichment + MD + FEP)		

5E — Uncertainty quantification & probabilistic ML (Weeks 54-55)

Week 54 · Sep 13-19 — Conformal prediction + calibration			
Day	Block A	Block B	Evening
Mon	Conformal prediction (Angelopoulos)	M: split conformal on your 5A model	Anki
Tue	Coverage guarantees; exchangeability	T: DP → empirical coverage check	Anki
Wed	MAPIE hands-on	M: MAPIE intervals + plot	Anki
Thu	Calibration: ECE, reliability, temp-scaling	T: graphs → reliability diagram	R: reproduce-paper
Fri	Scaffold/temporal shift breaks it	T: timed set	Whiteboard-Fri: why split conformal gives valid coverage

Sat-Sun	Buffer + rest		
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Week 55 · Sep 20-26 — GP / SVGP + UQ comparison (🕒 5E capstone)			
Day	Block A	Block B	Evening
Mon	GP regression: kernels, posterior	M: a GP on a small set (GPpyTorch)	Anki
Tue	SVGP (sparse variational) for scale	T: DP → SVGP on ADMET	Anki
Wed	Deep ensembles vs MC-dropout	M: ensemble + MC-dropout UQ	Anki
Thu	Four-way UQ comparison	T: graphs → conformal/GP/dropout/ensemble	R: write-up
Fri	Which to ship when	Wrap 5E capstone	Whiteboard-Fri: GP posterior + what the kernel encodes
Sat-Sun	Buffer + rest · <i>Deliverable:</i> 5E capstone (conformal + SVGP on 5A model; coverage)		

5F — Applied cheminformatics (Weeks 56-57)

Week 56 · Sep 27-Oct 3 — Retrosynthesis + reaction-condition + DEL			
Day	Block A	Block B	Evening
Mon	AIzynthFinder (MCTS retrosynthesis)	M: run retrosynthesis on public targets	Anki
Tue	Reaction representation + conditions	T: DP → reaction-condition model (USPTO/ORD)	Anki
Wed	DEL modeling; noisy counts	M: DEL enrichment on BELKA	Anki
Thu	Multitask / MMoE; contrastive	T: graphs → multi-endpoint ADMET	R: reproduce-paper
Fri	Synthesizability as a constraint	T: timed set	Whiteboard-Fri: MCTS retrosynthesis; DEL denoising
Sat-Sun	Buffer + rest		

Week 57 · Oct 4-10 — Federated + virtual screening + 🕒 5F capstone + track self-test			
Day	Block A	Block B	Evening (+ R)
Mon	Federated learning (Flower)	M: a federated multi-endpoint ADMET demo	Anki
Tue	Active-learning virtual screening	T: DP → AL loop (dock + surrogate)	Anki
Wed	Wrap the 5F capstone (pick a lane)	M: MCP-wrapped property+UQ tool (or chosen lane)	Anki
Thu	Consolidate the whole track	Polish repos + write-ups	R: post; submit one capstone to MLSB/LoG
Fri	🏁 PHASE-5 TRACK SELF-TEST (message-passing vs conv; GFlowNet objective; the accuracy/cost ladder; conformal coverage; SVGP vs dropout; MCTS/DEL/MMoE/federated)	Final polish	► Tier-2 applications (DESRES, Isomorphic, DeepMind-science)
Sat-Sun	Buffer + rest · <i>Deliverable:</i> 5F capstone (public)		

End of Phase 5 → Phase 6 (Scientific / chemistry agents) next. Tier 2 reached — drug-discovery credible.